

Problem List

53. Maximum Subarray

Medium

Topics

Companies

Given an integer array `nums`, find the **subarray** with the largest sum, and return its sum.

Example 1:

Input: `nums = [-2,1,-3,4,-1,2,1,-5,4]`
Output: 6
Explanation: The subarray `[4,-1,2,1]` has the largest sum 6.

Example 2:

Input: `nums = [1]`
Output: 1
Explanation: The subarray `[1]` has the largest sum 1.

Example 3:

Input: `nums = [5,4,-1,7,8]`
Output: 23
Explanation: The subarray `[5,4,-1,7,8]` has the largest sum 23.

Constraints:

- $1 \leq \text{nums.length} \leq 10^5$
- $-10^4 \leq \text{nums}[i] \leq 10^4$

Follow up: If you have figured out the $O(n)$ solution, try coding another solution using the **divide and conquer** approach, which is more subtle.

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Code

Java

Auto

```
1 class Solution {
2     public int maxSubArray(int[] nums) {
3         int maxSum = nums[0];
4         int currentSum = nums[0];
5
6         for (int i = 1; i < nums.length; i++) {
7             currentSum = Math.max(nums[i], currentSum + nums[i]);
8             maxSum = Math.max(maxSum, currentSum);
9         }
10
11         return maxSum;
12     }
13 }
```

Saved 1s 13 Col 2

Testcase

Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

nums =

[-2,1,-3,4,-1,2,1,-5,4]

Output

6

Expected

6

Contribute a testcase

Two children, Lily and Ron, want to share a chocolate bar. Each of the squares has an integer on it.

Lily decides to share a contiguous segment of the bar selected such that:

- The length of the segment matches Ron's birth month, and,
- The sum of the integers on the squares is equal to his birth day.

Determine how many ways she can divide the chocolate.

Example

$s = [2, 2, 1, 3, 2]$

$d = 4$

$m = 2$

Lily wants to find segments summing to Ron's birth day, $d = 4$ with a length equalling his birth month, $m = 2$. In this case, there are two segments meeting her criteria: $[2, 2]$ and $[1, 3]$.

Function Description

Complete the birthday function in the editor below.

birthday has the following parameter(s):

- `int s[n]`: the numbers on each of the squares of chocolate
- `int d`: Ron's birth day
- `int m`: Ron's birth month

Returns

```

1  import java.io.*;
2  import java.util.*;
3
4  public class Solution {
5
6      public static int birthday(List<Integer> s, int d, int m) {
7          int count = 0;
8
9          for (int i = 0; i <= s.size() - m; i++) {
10             int sum = 0;
11
12             for (int j = i; j < i + m; j++) {
13                 sum += s.get(j);
14             }
15
16             if (sum == d) {
17                 count++;
18             }
19         }
20
21         return count;
22     }
23
24     public static void main(String[] args) throws IOException {
25         BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
26
27         int n = Integer.parseInt(br.readLine().trim());
28     }
29 }

```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ **Sample Test case 0**

✓ Sample Test case 1

✓ Sample Test case 2

Input (stdin)

[Download](#)

1	5
2	1 2 1 3 2
3	3 2

Your Output (stdout)

1	2
---	---

Expected Output

[Download](#)

1	2
---	---

We define subsequence as any subset of an array. We define a subarray as a contiguous subsequence in an array.

Given an array, find the maximum possible sum among:

1. all nonempty subarrays.
2. all nonempty subsequences.

Print the two values as space-separated integers on one line.

Note that empty subarrays/subsequences should not be considered.

Example

$arr = [-1, 2, 3, -4, 5, 10]$

The maximum subarray sum is comprised of elements at indices $[1 - 5]$. Their sum is $2 + 3 + -4 + 5 + 10 = 16$. The maximum subsequence sum is comprised of elements at indices $[1, 2, 4, 5]$ and their sum is $2 + 3 + 5 + 10 = 20$.

Function Description

Complete the `maxSubarray` function in the editor below.

`maxSubarray` has the following parameter(s):

- `int arr[n]`: an array of integers

Returns

- `int[2]`: the maximum subarray and subsequence sums

Input Format

The first line of input contains a single integer t , the number of test cases.

The first line of each test case contains a single integer n .

The second line contains n space-separated integers $arr[i]$ where $0 \leq i < n$.

```
public class Solution {  
    public static List<Integer> maxSubarray(List<Integer> arr) {  
  
        // Maximum subsequence  
        if (x > 0) {  
            maxSubsequence += x;  
        }  
  
        // Largest element  
        if (x > maxElement) {  
            maxElement = x;  
        }  
    }  
  
    // If all elements are negative  
    if (maxSubsequence == 0) {  
        maxSubsequence = maxElement;  
    }  
  
    List<Integer> result = new ArrayList<Integer>();  
    result.add(maxSubarray);  
    result.add(maxSubsequence);  
  
    return result;  
}  
  
public static void main(String[] args) throws IOException {
```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

✔ Sample Test case 1

Input (stdin)

```
1 2
2 4
3 1 2 3 4
4 6
5 2 -1 2 3 4 -5
```

Your Output (stdout)

```
1 10 10
2 10 11
```

Expected Output

leetcod.com/problems/maximum-sum-circular-subarray/

Problem List

Run

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Description

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Example 1:

Input: nums = [1,-2,3,-2]

Output: 3

Explanation: Subarray [3] has maximum sum 3.

Example 2:

Input: nums = [5,-3,5]

Output: 10

Explanation: Subarray [5,5] has maximum sum 5 + 5 = 10.

Example 3:

Input: nums = [-3,-2,-3]

Output: -2

Explanation: Subarray [-2] has maximum sum -2.

Constraints:

n == nums.length

1 <= n <= 3 * 10⁴

-3 * 10⁴ <= nums[i] <= 3 * 10⁴

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Seen this question in a real interview before?

1/6

Yes

No

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Acceptance Rate 51.3%

Topics

7.5K

135

46 Online

Code

Note

Java

Auto

12

13

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33

34

for (int num : nums) {

// Maximum subarray sum (Kadane's algorithm)

currentMax = Math.max(num, currentMax + num);

maxSum = Math.max(maxSum, currentMax);

// Minimum subarray sum

currentMin = Math.min(num, currentMin + num);

minSum = Math.min(minSum, currentMin);

total += num;

}

// If all elements are negative

if (maxSum < 0) {

return maxSum;

}

// Circular maximum = total sum - minimum subarray sum

int circularSum = total - minSum;

return Math.max(maxSum, circularSum);

}

Saved

Ln 34, Col 3

Testcase

Test Result

Accepted

Runtime: 0 ms

Case 1

Case 2

Case 3

Input

nums =

[1, -2, 3, -2]

Output

34°C

Mostly sunny

Search

ENG US

11:45

28-08-2026

String to Integer (atoi) - LeetC...Maximum Sum Circular Subarr...Contains Duplicate Solution

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Verify it's you

Problem List

Submit

8. String to Integer (atoi)

MediumTopicsCompanies

Implement the `myAtoi(string s)` function, which converts a string to a 32-bit signed integer.

The algorithm for `myAtoi(string s)` is as follows:

1. **Whitespace:** Ignore any leading whitespace (" ").

2. **Signedness:** Determine the sign by checking if the next character is '-' or '+', assuming positivity if neither present.

3. **Conversion:** Read the integer by skipping leading zeros until a non-digit character is encountered or the end of the string is reached. If no digits were read, then the result is 0.

4. **Rounding:** If the integer is out of the 32-bit signed integer range $[-2^{31}, 2^{31} - 1]$, then round the integer to remain in the range. Specifically, integers less than -2^{31} should be rounded to -2^{31} , and integers greater than $2^{31} - 1$ should be rounded to $2^{31} - 1$.

Return the integer as the final result.

Example 1:

Input: s = "42"

Output: 42

Explanation:

The underlined characters are what is read in and the caret is the current reader position.

Step 1: "42" (no characters read because there is no leading whitespace)

Step 2: "42" (no characters read because there is neither a '-' nor '+')

Step 3: "42" ("42" is read in)

Example 2:

Input: s = "-042"

6.4K992159 Online

CodeNote

JavaAuto

```
15 if (i < n && s.charAt(i) == '-') {
16     sign = -1;
17     i++;
18 } else if (i < n && s.charAt(i) == '+') {
19     i++;
20 }
21
22 // 3. Convert digits
23 int result = 0;
24
25 while (i < n && Character.isDigit(s.charAt(i))) {
26     int digit = s.charAt(i) - '0';
27
28     // 4. Check for overflow before adding digit
29     if (result > (Integer.MAX_VALUE - digit) / 10) {
30         return sign == 1 ? Integer.MAX_VALUE : Integer.MIN_VALUE;
31     }
32
33     result = result * 10 + digit;
34     i++;
35 }
36
37 return result * sign;
38 }
39 }
```

SavedLn 39, Col 2

Testcase>Test Result

AcceptedRuntime: 0 ms

Case 1Case 2Case 3Case 4Case 5

Input

s =

"42"

Output

34°C
Mostly sunny

Search

ENG
US

11:48
28-08-2026

You are given a string containing characters A and B only. Your task is to change it into a string such that there are no matching adjacent characters. To do this, you are allowed to delete zero or more characters in the string.

Your task is to find the minimum number of required deletions.

Example

$s = AABAAB$

Remove an A at positions 0 and 3 to make $s = ABAB$ in 2 deletions.

Function Description

Complete the `alternatingCharacters` function in the editor below.

`alternatingCharacters` has the following parameter(s):

- string s : a string

Returns

- int: the minimum number of deletions required

Input Format

The first line contains an integer q , the number of queries.

The next q lines each contain a string s to analyze.

Constraints

- $1 \leq q \leq 10$
- $1 \leq \text{length of } s \leq 10^5$
- Each string s will consist only of characters A and B .

Sample Input

```
1  import java.io.*;
2  import java.util.*;
3
4  public class Solution {
5
6      public static int alternatingCharacters(String s) {
7          int deletions = 0;
8
9          for (int i = 1; i < s.length(); i++) {
10             if (s.charAt(i) == s.charAt(i - 1)) {
11                 deletions++;
12             }
13         }
14
15         return deletions;
16     }
17
18     public static void main(String[] args) throws IOException {
19         BufferedReader br = new BufferedReader(
20             new InputStreamReader(System.in)
21         );
22
23         int q = Integer.parseInt(br.readLine().trim());
24
25         for (int i = 0; i < q; i++) {
26             String s = br.readLine().trim();
27             System.out.println(alternatingCharacters(s));
28         }
29     }
30 }
```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

✔ Sample Test case 1

✔ Sample Test case 2

Input (stdin)

1	5
2	AAAA
3	BBBBB
4	ABABABAB
5	BABABA
6	AAABBB

Your Output (stdout)

1	3
2	4
3	0
4	0

3. Longest Substring Without Repeating Characters

Medium Topics Companies Hint

Given a string s, find the length of the longest substring without duplicate characters.

Example 1:

Input: s = "abcabcbb"
Output: 3
Explanation: The answer is "abc", with the length of 3. Note that "bca" and "cab" are also correct answers.

Example 2:

Input: s = "bbbbb"
Output: 1
Explanation: The answer is "b", with the length of 1.

Example 3:

Input: s = "pwwkew"
Output: 3
Explanation: The answer is "wke", with the length of 3. Notice that the answer must be a substring, "pwke" is a subsequence and not a substring.

Constraints:

- 0 ≤ s.length ≤ 10⁵
- s consists of English letters, digits, symbols and spaces.

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```
14         char ch = s.charAt(right);
15
16         // If character was already seen inside the window
17         if (lastIndex[ch] >= left) {
18             left = lastIndex[ch] + 1;
19         }
20
21         lastIndex[ch] = right;
22
23         maxLength = Math.max(maxLength, right - left + 1);
24     }
25
26     return maxLength;
27 }
28
29 }
```

Saved

Ln 2

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

s = "abcabcbb"

Output

3

Expected

3

890. Find and Replace Pattern

Medium Topics Companies

Given a list of strings `words` and a string `pattern`, return a list of `words[i]` that match `pattern`. You may return the answer in **any order**.

A word matches the pattern if there exists a permutation of letters `p` so that after replacing every letter `x` in the pattern with `p(x)`, we get the desired word.

Recall that a permutation of letters is a bijection from letters to letters: every letter maps to another letter, and no two letters map to the same letter.

Example 1:

Input: `words = ["abc","deq","mee","aqq","dkd","ccc"], pattern = "abb"`

Output: `["mee","aqq"]`

Explanation: "mee" matches the pattern because there is a permutation {a -> m, b -> e, ...}. "ccc" does not match the pattern because {a -> c, b -> c, ...} is not a permutation, since a and b map to the same letter.

Example 2:

Input: `words = ["a","b","c"], pattern = "a"`

Output: `["a","b","c"]`

Constraints:

- `1 <= pattern.length <= 20`
- `1 <= words.length <= 50`
- `words[i].length == pattern.length`
- `pattern` and `words[i]` are lowercase English letters.

4K 28 ☆ 10 Online

```
31:         }
32:
33:         // Check word -> pattern mapping
34:         if (wordToPattern.containsKey(w)) {
35:             if (wordToPattern.get(w) != p) {
36:                 return false;
37:             }
38:         } else {
39:             wordToPattern.put(w, p);
40:         }
41:     }
42:
43:     return true;
44: }
45:
46:
```

Saved

Testcase | >_ Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input

words =
["abc","deq","mee","aqq","dkd","ccc"]

pattern =
"abb"

Output

["mee","aqq"]

Expected

["mee","aqq"]

Problem List

1408. String Matching in an Array

EasyTopicsCompaniesHint

Given an array of string words, return all strings in words that are a substring of another word. You can return the answer in any order.

Example 1:
Input: words = ["mass","as","hero","superhero"]
Output: ["as","hero"]
Explanation: "as" is substring of "mass" and "hero" is substring of "superhero". ["hero","as"] is also a valid answer.

Example 2:
Input: words = ["leetcode","et","code"]
Output: ["et","code"]
Explanation: "et", "code" are substring of "leetcode".

Example 3:
Input: words = ["blue","green","bu"]
Output: []
Explanation: No string of words is substring of another string.

Constraints:

- 1 <= words.length <= 100
- 1 <= words[i].length <= 30
- words[i] contains only lowercase English letters.
- All the strings of words are unique.

1.5K17310 Online

Code

Java

Auto

1import java.util.*;
2
3class Solution {
4 public List<String> stringMatching(String[] words) {
5 List<String> result = new ArrayList<>();
6
7 for (int i = 0; i < words.length; i++) {
8 for (int j = 0; j < words.length; j++) {
9 if (i != j && words[j].contains(words[i])) {
10 result.add(words[i]);
11 break;
12 }
13 }
14 }
15
16 return result;
17 }
18 }

SavedLn 18, Col 2

Testcase>Test Result

AcceptedRuntime: 0 ms

Case 1Case 2Case 3

Input
words =
["mass","as","hero","superhero"]

Output

34°C
Mostly sunny

Search

ENG
US

11:31
28-08-2026

WhatsApp

Naive algorithm for Pattern Se...

String Matching in an Array - L...

String to Integer (atoi) - LeetCo...

Maximum Sum Circular Subarr...

Contains Duplicate Solution

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geeksforgeeks.org/dsa/naive-algorithm-for-pattern-searching/

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```
int j;

// Check for pattern match at index i
for (j = 0; j < m; j++)
{
    if (txt.charAt(i + j) != pat.charAt(j))
        break;
}

// If pattern matches, store index
if (j == m)
    ans.add(i);
}

return ans;
}

public static void main(String[] args)
{
    String txt = "aabaacaadaabaaba";
    String pat = "aaba";

    ArrayList<Integer> res = search(pat, txt);

    for (int it : res)
    {
        System.out.print(it + " ");
    }
}
```

Output

0 9 12

Best Case Scenario

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